KAIZEN MATHS

Lesson starter

Trigonometric Equation Solver

Quadratic Trig Equations

Name _____

S1

Solve $2\sin^2(3x+\frac{\pi}{3})-\sin(3x+\frac{\pi}{3})-1=0$ for $0\le x<2\pi$.

S2

Solve $2\cos^2(3x+\frac{\pi}{3})+\cos(3x+\frac{\pi}{3})-1=0$ for $0\le x<2\pi$.

S3

Solve $2\sin^2(3x+\frac{\pi}{4})-\sin(3x+\frac{\pi}{4})-1=0$ for $0\le x<2\pi$.

S4

Solve $2\cos^2(2x+\frac{\pi}{4})+\cos(2x+\frac{\pi}{4})-1=0$ for $0\le x<2\pi$.

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Solve $2\sin^2(3x+\frac{\pi}{4})-\sin(3x+\frac{\pi}{4})-1=0$ for $0\le x<2\pi$.

S4

Solve $2\cos^2(2x+\frac{\pi}{4})+\cos(2x+\frac{\pi}{4})-1=0$ for $0\le x<2\pi$.

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KAIZEN MATHS

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Trigonometric Equation Solver

Quadratic Trig Equations

Name _____

E1

Solve $2\sin^2(2x+\frac{\pi}{6})-\sin(2x+\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

E2

Solve $2\cos^2(2x-\frac{\pi}{6})+\cos(2x-\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

E3

Solve $2\sin^2(3x-\frac{\pi}{3})-\sin(3x-\frac{\pi}{3})-1=0$ for $0\le x<2\pi$.

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Trigonometric Equation Solver

Quadratic Trig Equations

Name _____

E1

Solve $2\sin^2(2x+\frac{\pi}{6})-\sin(2x+\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

E2

Solve $2\cos^2(2x-\frac{\pi}{6})+\cos(2x-\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

E3

Solve $2\sin^2(3x-\frac{\pi}{3})-\sin(3x-\frac{\pi}{3})-1=0$ for $0\le x<2\pi$.

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Trigonometric Equation Solver

Quadratic Trig Equations

Name _____

E1

Solve $2\sin^2(2x+\frac{\pi}{6})-\sin(2x+\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

E2

Solve $2\cos^2(2x-\frac{\pi}{6})+\cos(2x-\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

E3

Solve $2\sin^2(3x-\frac{\pi}{3})-\sin(3x-\frac{\pi}{3})-1=0$ for $0\le x<2\pi$.

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Trigonometric Equation Solver

Quadratic Trig Equations

Name _____

E1

Solve $2\sin^2(2x+\frac{\pi}{6})-\sin(2x+\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

E2

Solve $2\cos^2(2x-\frac{\pi}{6})+\cos(2x-\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

E3

Solve $2\sin^2(3x-\frac{\pi}{3})-\sin(3x-\frac{\pi}{3})-1=0$ for $0\le x<2\pi$.

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Teacher answers & working

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S1

Solve $2\sin^2(3x+\frac{\pi}{3})-\sin(3x+\frac{\pi}{3})-1=0$ for $0\le x<2\pi$.

Answer:

$x=\frac{\pi}{6},1.22,1.92,\frac{5\pi}{6},3.32,4.01,\frac{3\pi}{2},5.41,6.11.$

Working:

Let $\theta=3x+\frac{\pi}{3}$.

Factor the quadratic: $(2\sin(\theta)-1)(\sin(\theta)+1)=0$.

So $\sin(\theta)=\frac{1}{2}$ or $\sin(\theta)=-1$.

This gives $\theta=\frac{\pi}{6},\frac{5\pi}{6},\frac{3\pi}{2}$.

Solve $3x+\frac{\pi}{3}=\theta$ to get $x=\frac{\pi}{6},1.22,1.92,\frac{5\pi}{6},3.32,4.01,\frac{3\pi}{2},5.41,6.11$.

S2

Solve $2\cos^2(3x+\frac{\pi}{3})+\cos(3x+\frac{\pi}{3})-1=0$ for $0\le x<2\pi$.

Answer:

$x=0,0.70,1.40,\frac{2\pi}{3},2.79,3.49,\frac{4\pi}{3},4.89,5.59.$

Working:

Let $\theta=3x+\frac{\pi}{3}$.

Factor the quadratic: $(2\cos(\theta)-1)(\cos(\theta)+1)=0$.

So $\cos(\theta)=\frac{1}{2}$ or $\cos(\theta)=-1$.

This gives $\theta=\frac{\pi}{3},\pi,\frac{5\pi}{3}$.

Solve $3x+\frac{\pi}{3}=\theta$ to get $x=0,0.70,1.40,\frac{2\pi}{3},2.79,3.49,\frac{4\pi}{3},4.89,5.59$.

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S3

Solve $2\sin^2(3x+\frac{\pi}{4})-\sin(3x+\frac{\pi}{4})-1=0$ for $0\le x<2\pi$.

Answer:

$x=0.61,\frac{5\pi}{12},2.01,2.71,\frac{13\pi}{12},4.10,4.80,\frac{7\pi}{4},6.20.$

Working:

Let $\theta=3x+\frac{\pi}{4}$.

Factor the quadratic: $(2\sin(\theta)-1)(\sin(\theta)+1)=0$.

So $\sin(\theta)=\frac{1}{2}$ or $\sin(\theta)=-1$.

This gives $\theta=\frac{\pi}{6},\frac{5\pi}{6},\frac{3\pi}{2}$.

Solve $3x+\frac{\pi}{4}=\theta$ to get $x=0.61,\frac{5\pi}{12},2.01,2.71,\frac{13\pi}{12},4.10,4.80,\frac{7\pi}{4},6.20$.

S4

Solve $2\cos^2(2x+\frac{\pi}{4})+\cos(2x+\frac{\pi}{4})-1=0$ for $0\le x<2\pi$.

Answer:

$x=0.13,1.18,2.23,3.27,4.32,5.37.$

Working:

Let $\theta=2x+\frac{\pi}{4}$.

Factor the quadratic: $(2\cos(\theta)-1)(\cos(\theta)+1)=0$.

So $\cos(\theta)=\frac{1}{2}$ or $\cos(\theta)=-1$.

This gives $\theta=\frac{\pi}{3},\pi,\frac{5\pi}{3}$.

Solve $2x+\frac{\pi}{4}=\theta$ to get $x=0.13,1.18,2.23,3.27,4.32,5.37$.

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E1

Solve $2\sin^2(2x+\frac{\pi}{6})-\sin(2x+\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

Answer:

$x=0,\frac{\pi}{3},\frac{2\pi}{3},\pi,\frac{4\pi}{3},\frac{5\pi}{3}$.

Working:

Let $\theta=2x+\frac{\pi}{6}$.

Factor the quadratic: $(2\sin(\theta)-1)(\sin(\theta)+1)=0$.

So $\sin(\theta)=\frac{1}{2}$ or $\sin(\theta)=-1$.

This gives $\theta=\frac{\pi}{6},\frac{5\pi}{6},\frac{3\pi}{2}$.

Solve $2x+\frac{\pi}{6}=\theta$ to get $x=0,\frac{\pi}{3},\frac{2\pi}{3},\pi,\frac{4\pi}{3},\frac{5\pi}{3}$.

E2

Solve $2\cos^2(2x-\frac{\pi}{6})+\cos(2x-\frac{\pi}{6})-1=0$ for $0\le x<2\pi$.

Answer:

$x=\frac{\pi}{4},\frac{7\pi}{12},\frac{11\pi}{12},\frac{5\pi}{4},\frac{19\pi}{12},\frac{23\pi}{12}$.

Working:

Let $\theta=2x-\frac{\pi}{6}$.

Factor the quadratic: $(2\cos(\theta)-1)(\cos(\theta)+1)=0$.

So $\cos(\theta)=\frac{1}{2}$ or $\cos(\theta)=-1$.

This gives $\theta=\frac{\pi}{3},\pi,\frac{5\pi}{3}$.

Solve $2x-\frac{\pi}{6}=\theta$ to get $x=\frac{\pi}{4},\frac{7\pi}{12},\frac{11\pi}{12},\frac{5\pi}{4},\frac{19\pi}{12},\frac{23\pi}{12}$.

Teacher answers & working

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E3

Solve $2\sin^2(3x - \frac{\pi}{3}) - \sin(3x - \frac{\pi}{3}) - 1 = 0$ for $0 \leq x < 2\pi$.

Answer:

$x = \frac{\pi}{6}, 1.22, 1.92, \frac{5\pi}{6}, 3.32, 4.01, \frac{3\pi}{2}, 5.41, 6.11.$

Working:

Let $\theta = 3x - \frac{\pi}{3}$.

Factor the quadratic: $(2\sin(\theta) - 1)(\sin(\theta) + 1) = 0$.

So $\sin(\theta) = \frac{1}{2}$ or $\sin(\theta) = -1$.

This gives $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$.

Solve $3x - \frac{\pi}{3} = \theta$ to get $x = \frac{\pi}{6}, 1.22, 1.92, \frac{5\pi}{6}, 3.32, 4.01, \frac{3\pi}{2}, 5.41, 6.11.$